

DEPLOYMENT_GUIDE

HelixTrack Deployment Guide — 2026-06-21

Overview

Deployment guide for HelixTrack across all platforms.

Prerequisites

Development

- Go 1.22+
- Node.js 18+
- Angular CLI 19
- Docker + Docker Compose
- Android Studio (for Android)
- Xcode (for iOS)
- Rust + Tauri CLI (for Desktop)

Production

- PostgreSQL 12+
 - Redis (optional, for caching)
 - Nginx (reverse proxy)
 - SSL certificate
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Backend Deployment

Development

```
cd core/Application  
go run main.go
```

Docker

```
cd containers
docker compose -f compose.helixtrack.yml up -d
```

Production

```
# Build
cd core/Application
go build -o htCore main.go

# Run
./htCore --space-root=/data/spaces/_default
```

Web Client Deployment

Development

```
cd web_client
npm install
npm start
# Access at http://localhost:4200
```

Production Build

```
cd web_client
npm run build --configuration=production
# Output in dist/helixtrack-client/
```

Docker

```
# Build image
docker build -t helixtrack-web .

# Run
docker run -p 80:80 helixtrack-web
```

Desktop Client Deployment

Development

```
cd desktop_client
npm install
npm run tauri dev
```

Build

```
cd desktop_client
npm run tauri build
# Output in src-tauri/target/release/
```

Platforms

- macOS: .dmg
 - Windows: .msi
 - Linux: .AppImage
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Android Client Deployment

Development

```
cd android_client
./gradlew installDebug
```

Build

```
cd android_client
./gradlew assembleRelease
# Output in app/build/outputs/apk/
```

Play Store

1. Update version in build.gradle
 2. Generate signed APK
 3. Upload to Play Console
-

iOS Client Deployment

Development

```
cd ios_client
open HelixTrack.xcodeproj
# Run on simulator
```

Build

```
cd ios_client
xcodebuild -scheme HelixTrack -configuration Release
```

App Store

1. Update version in Xcode
 2. Archive
 3. Upload to App Store Connect
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Database Setup

SQLite (Development)

```
# Auto-created on first run  
# Location: core/Application/Database/Definition.sqlite
```

PostgreSQL (Production)

```
CREATE DATABASE helixtrack;  
CREATE USER helixtrack WITH PASSWORD 'your_password';  
GRANT ALL PRIVILEGES ON DATABASE helixtrack TO helixtrack;
```

Environment Variables

Backend

```
HT_CONFIG=/path/to/config.json  
HT_SPACE_ROOT=/data/spaces/_default  
HT_DB_TYPE=sqlite # or postgres  
HT_PG_PASSWORD=your_password
```

Web Client

```
NG_APP_API_URL=http://localhost:8080  
NG_APP_WS_URL=ws://localhost:8080/ws
```

Monitoring

Health Check

```
curl http://localhost:8080/health
```

Logs

```
# Backend
docker logs helixtrack-core

# Web Client
# Check browser console
```

Backup

Database

```
# SQLite
cp Definition.sqlite Definition.sqlite.backup

# PostgreSQL
pg_dump helixtrack > backup.sql
```

Files

```
# Spaces data
tar -czf spaces-backup.tar.gz /data/spaces/
```

Cross-references

- [Architecture](#)
- [API Reference](#)

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/ (sibling HelixTrack source tree) , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced HelixTrack source tree + upstream umbrella, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - The deployment steps describe the sibling HelixTrack project (Core backend + Web/Desktop clients confirmed present locally and in the github.com/Helix-Track/Everything umbrella). - **Negative finding:** deployment-platform-specific commands (any third-party runtime/CLI versions invoked) were NOT re-verified against their respective upstream docs in this pass — only the HelixTrack source structure was confirmed. Before relying on any external-tool command here (container runtime, package manager), re-verify against that tool's latest official docs per §11.4.99. The backup/data paths are illustrative of the project layout.